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Consider an object thrown with initial velocity $\vec{u}$ at an angle θ with the horizontal. Refer Figures 2.42 and 2.43.

Then,

$$\vec{u} = u_x \hat{i} + u_y \hat{j}$$

where $u_x = u\cos\theta$ is the horizontal component and $u_y = usin\theta$ the vertical component of velocity.

Since the acceleration due to gravity is in the direction opposite to the direction of vertical component u_y, this component will gradually reduce to zero at the maximum height of the projectile. At this maximum height, the same gravitational force will push the projectile to move downward and fall to the ground. There is no acceleration along the x direction throughout the motion. So, the horizontal component of the velocity ($u_x = u\cos\theta$) remains the same till the object reaches the ground.

Hence after the time t, the velocity along horizontal motion $v_x = u_x + a_x t = u_x = u\cos\theta$

The horizontal distance travelled by projectile in time t is $s_x = u_x t + \frac{1}{2} a_x t^2$

Here, $s_x = x$, $u_x = u\,cos\theta$, $a_x = 0$

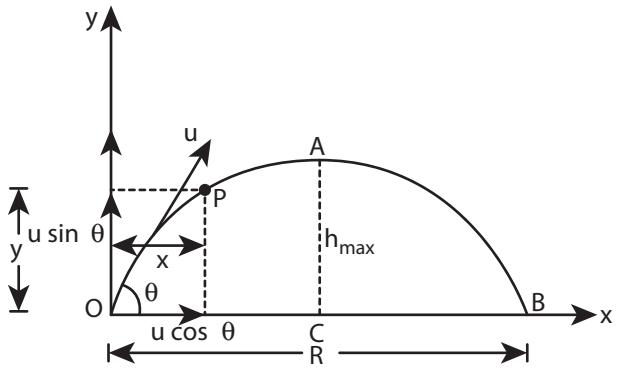


Figure 2.43. Initial velocity resolved into components

$$\text{Thus, } x = u\cos\theta.t \text{ or } t = \frac{x}{u\cos\theta} \quad (2.28)$$

Next, for the vertical motion $v_y = u_y + a_y t$

Here $u_y = u\,sin\theta$, $a_y = -g$ (acceleration due to gravity acts opposite to the motion). Thus

$$\text{Thus, } v_y = usin\theta - gt \quad (2.29)$$

The vertical distance travelled by the projectile in the same time t is $s_y = u_y t + \frac{1}{2} a_y t^2$

Here, $s_y = y$, $u_y = u\,sin\theta$, $a_x = -g$. Then

$$y = u\,\sin\theta\, t - \frac{1}{2} g t^2 \quad (2.30)$$

Substitute the value of t from equation (2.28) in equation (2.30), we have

$$y = u\sin\theta \frac{x}{u\cos\theta} - \frac{1}{2} g \frac{x^2}{u^2\cos^2\theta}$$

$$y = x\tan\theta - \frac{1}{2} g \frac{x^2}{u^2\cos^2\theta} \quad (2.31)$$

Thus the path followed by the projectile is an inverted parabola.

Maximum height (h_{max})

The maximum vertical distance travelled by the projectile during its journey is called maximum height. This is determined as follows:

For the vertical part of the motion,

$$v_y^2 = u_y^2 + 2a_y s$$

Here, $u_y = u\,sin\theta$, $a = -g$, $s = h_{max}$, and at the maximum height $v_y = 0$